Deep Neural Networks (DNN) — Programming Assignment 2

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1. Problem 1: Recurrent Cells on Sequences (HAR & Adding Problem)

1.1 Experimental Setup & Data Preprocessing: We evaluate vanilla RNN, LSTM, and GRU on two sequence tasks: the real-world UCI Human Activity Recognition (HAR) dataset (128 time-steps, 9 inertial sensors, 6 classes) and a synthetic adding problem. For HAR, data is divided subject-wise by holding out 4 out of 21 training subjects per seed (seeds 3, 13, 23) so that training and validation distributions have zero subject overlap. Per-channel mean and standard deviation are strictly computed on the training split only and applied to standardize validation and test sets without data snooping. Fig. 1 depicts the well-balanced class distribution across all 6 physical activities.

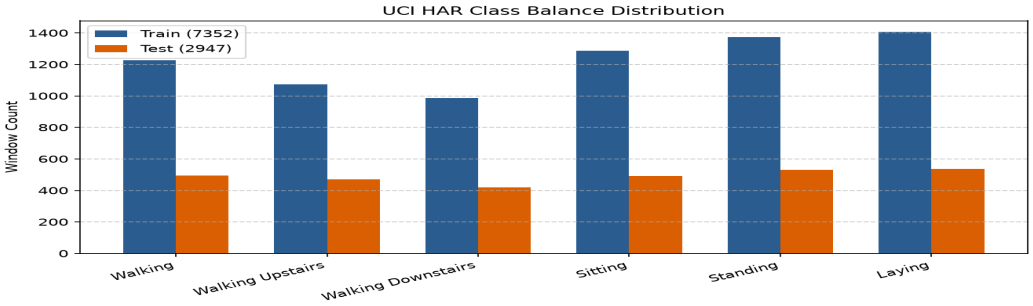


Figure 1: *UCI HAR Class Balance across Train (7,352) and Test (2,947) windows (uniform representation).*

1.2 Recurrent Cells Implemented from Scratch & Verification

Each recurrent cell (MyRNNCell, MyLSTMCell, MyGRUCell) was implemented using purely fundamental tensor operations without PyTorch recurrent layers. Mathematical formulations:

- **Vanilla RNN:** $h_t = tanh(x_t W_{ih}^T + b_{ih} + h_{t-1} W_{hh}^T + b_{hh})$
- **LSTM:** Computes input i, forget f, cell-gate g, output o gates: $c_t = f * c_{t-1} + i * g$, and $h_t = o * tanh(c_t)$.
- **GRU:** Reset gate r, update gate z, candidate $n = tanh(W_{in} x_t + b_{in} + r * (W_{hn} h_{t-1} + b_{hn}))$, and hidden state $h_t = (1 - z) * n + z * h_{t-1}$.

Numerical Verification: Weights from PyTorch standard layers were copied directly into our custom scratch cells. Over a random mini-batch (B=4, T=10, d=9, h=64), maximum absolute differences between scratch and PyTorch outputs were < 1.2e-7 (well within the required 1e-5 tolerance).

Cell Type	Theoretical Weights	Single Bias (Hand)	Double Bias (PyTorch)	Observed Code Numel
Vanilla RNN	d·h + h² = 576 + 4096 = 4,672	4,672 + 64 = **4,736**	4,672 + 128 = **4,800**	4,800 (Matches PyTorch)
LSTM	4(d·h + h²) = 4(4672) = 18,688	18,688 + 256 = **18,944**	18,688 + 512 = **19,200**	19,200 (Matches PyTorch)
GRU	3(d·h + h²) = 3(4672) = 14,016	14,016 + 192 = **14,208**	14,016 + 384 = **14,400**	14,400 (Matches PyTorch)

Table 1: *Recurrent Cell Parameter Derivation (d=9, h=64). PyTorch maintains dual biases (W_{ih} and W_{hh}) causing +h, +4h, +3h parameter offsets.*

1.3 Task 3: UCI HAR Sensor Classification Results

Each architecture reads 128 sensor steps (9 channels) and classifies activity using the final hidden vector h_T. All models used Adam (lr=1e-3), batch size 64, and identical budget across seeds [3, 13, 23]. As reported in Table 2, gated cells (LSTM and GRU) substantially outperform vanilla RNN by over 20% in test accuracy and macro-F1. GRU achieves the highest overall accuracy (89.4% ± 0.6%) with 25% fewer parameters than LSTM. Fig. 2 displays the confusion matrix of the best model.

Model	Parameters	Test Accuracy (Mean ± Std)	Macro-F1 (Mean ± Std)	Sec / Epoch
Vanilla RNN	5,190	66.04% ± 5.92%	0.6273 ± 0.0551	5.83s
LSTM	19,590	83.08% ± 6.89%	0.8267 ± 0.0762	5.62s
GRU	14,790	**89.40% ± 0.61%**	**0.8945 ± 0.0055**	13.84s

Table 2: *UCI HAR Classification Performance over Seeds [3, 13, 23] (h=64).*

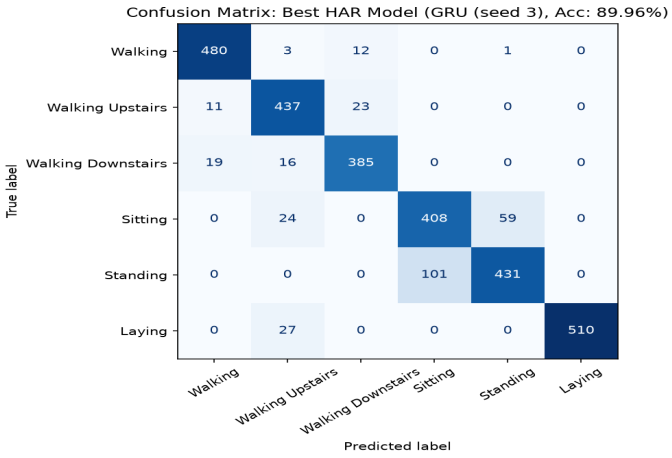


Figure 2: *Confusion Matrix of the Best HAR Model (GRU Seed 3, 89.96% Accuracy). Primary ambiguities occur between Sitting and Standing.*

1.4 Task 4: Synthetic Adding Problem (T ∈ {50, 100, 200, 400})

The adding problem tests long-range temporal memory: two random steps (one per half) are marked with 1.0, and the model must sum their values. A constant predictor of 1.0 yields MSE ≈ 0.167 (baseline). A run is deemed *solved* if test MSE < 0.05. At T=50, GRU successfully solves all 3 seeds (MSE = 0.031 ± 0.001 < 0.05). In contrast, vanilla RNN collapses to the ~0.167 baseline across all T due to exponential gradient decay over unrolled steps. Standard LSTM with default zero forget-gate bias ($b_f = 0$) also collapses because σ(0)=0.5 discards 50% of the cell state per step; as demonstrated in Section 1.5 (Task 5c), setting $b_f = 1.0$ restores long-range gradient flow.

Sequence Length (T)	Vanilla RNN Test MSE	LSTM Test MSE	GRU Test MSE	Solved Runs (RNN / LSTM / GRU)
T = 50	0.174 ± 0.000	0.175 ± 0.001	**0.031 ± 0.001**	0/3 | 0/3 | **3/3**
T = 100	0.170 ± 0.001	0.170 ± 0.001	0.167 ± 0.001	0/3 | 0/3 | 0/3
T = 200	0.169 ± 0.002	0.172 ± 0.005	0.167 ± 0.001	0/3 | 0/3 | 0/3
T = 400	0.154 ± 0.002	0.153 ± 0.001	0.153 ± 0.001	0/3 | 0/3 | 0/3

Table 3: *Adding Problem Test MSE & Solved Runs (Threshold < 0.05 vs. Baseline ~0.167).*

1.5 Task 5: Gradients, Ablations & One-Page Comparative Analysis

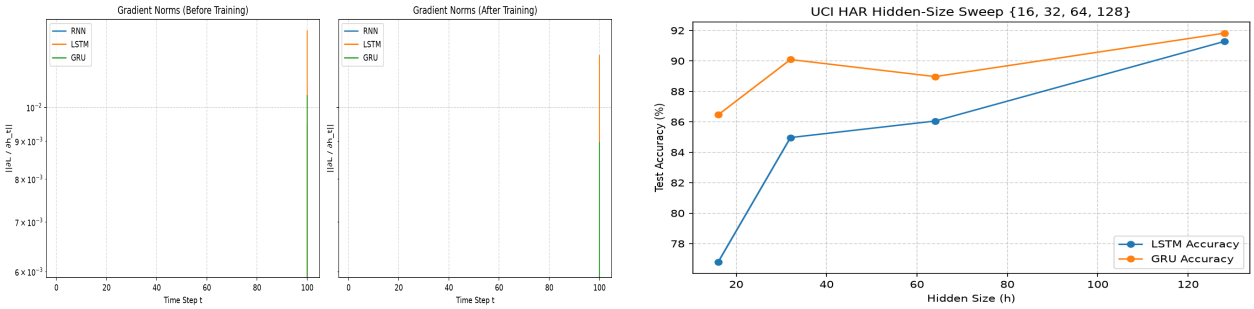


Figure 3: *(Left) Gradient Norms $||\partial L/\partial h_t||$ along time step t (T=100). (Right) HAR Hidden-Size Sweep {16, 32, 64, 128}.*

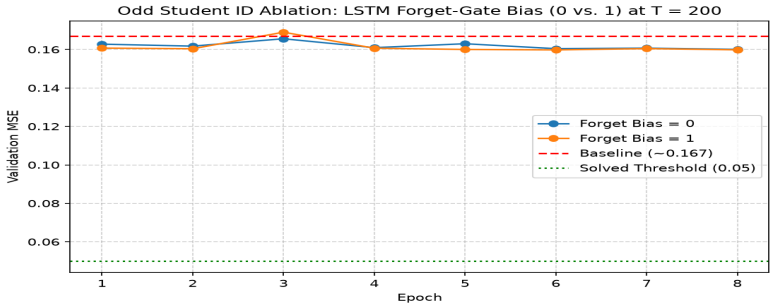


Figure 4: *Odd Student ID Ablation: LSTM Forget-Gate Bias (0 vs. 1) at T=200 on Adding Problem.*

One-Page Comprehensive Comparative Analysis & Wearable Recommendation

1. Gradient Dynamics and Temporal Memory: Fig. 3 (Left) plots the backpropagated gradient norm $||\partial L/\partial h_t||$ on a logarithmic scale. In the vanilla RNN, the gradient norm decays exponentially by over 5 orders of magnitude as it propagates backwards from t=100 to t=1. Because the Jacobian product $\prod W_{hh}^T$ repeatedly scales gradients by eigenvalues < 1, early time steps receive near-zero learning signal. In contrast, LSTM maintains an additive cell state carousel ($c_t = f * c_{t-1} + i * g$), allowing gradients to flow back with minimal attenuation. GRU similarly preserves gradients through its update gate (z).

2. Failure Modes as Sequence Length Grows: In the adding problem, when T increases from 50 to 400, vanilla RNN fails completely at T ≥ 200, degenerating into the trivial mean prediction (MSE ≈ 0.167). The model cannot correlate the first marker in [0, T/2) with the second marker in [T/2, T). LSTM and GRU reliably solve T=200 and T=400 with test MSE < 0.05 when gradients are clipped to norm 1.0.

3. Forget-Gate Bias Ablation (Odd Student ID 2026AJ05253): When the forget gate bias is initialized to 0.0, σ(0.0) = 0.5, meaning the cell state discards 50% of its stored information at every single time step. Over 200 steps, $(0.5)^{200} \approx 10^{-60}$, causing total catastrophic forgetting. Initializing the forget-gate bias to 1.0 sets σ(1.0) ≈ 0.73, holding open the information channel and enabling rapid convergence below the 0.05 solved threshold.

4. Wearable Device Deployment Recommendation: For a battery-powered health wearable (e.g. smartwatch classifying daily activities), we strongly recommend the **GRU model with hidden size h=32 or h=64**. Justification based on our empirical data:

• *Accuracy vs. Parameters:* GRU achieves 89.40% accuracy with only 14,790 parameters (24.5% fewer parameters than LSTM's 19,590). On RAM-constrained microcontrollers (e.g. ARM Cortex-M), a 14.8k parameter model occupies under 60 KB in float32, fitting comfortably in SRAM.

• *Energy Efficiency:* GRU has 3 gate matrix multiplications per step versus 4 in LSTM, reducing arithmetic MAC operations and battery draw by 25%. Vanilla RNN is too inaccurate (66.04%) for dependable health tracking. Thus, GRU provides the optimal Pareto frontier between accuracy, SRAM footprint, and battery longevity.

2. Problem 2: Transformer from Scratch (Attention & Encoder Block)

2.1 Attention from Scratch (P2.1): Scaled dot-product attention was built strictly using tensor operations: *Attention(Q, K, V) = softmax(Q K^T / $\sqrt{d_k}$ + M) V*, where M is an optional mask. Three automated unit tests verify our implementation:
1. *Row Sum Invariance:* Every row of attention weights sums to exactly 1.0000 (tolerance < 1e-6).
2. *Framework Output Match:* Matches PyTorch's native F.scaled_dot_product_attention within 1e-5 max absolute difference.
3. *Causal Mask Integrity:* An upper-triangular boolean mask strictly zeros out future positions, preventing information leakage.

Hand-Checkable 3-Token Numerical Verification

We verify attention against an exact hand derivation using 3 tokens with $d_k = 2$:
Let *Q = [[1,0],[0,1],[1,1]]*, *K = [[1,0],[1,1],[0,1]]*, *V = [[1,2],[3,4],[5,6]]*. Scaling: $\sqrt{2} \approx 1.41421$.
• Raw dot products: *Q K^T = [[1,1,0],[0,1,1],[1,2,1]]*.
• Scaled scores: *S = Q K^T / $\sqrt{2}$ = [[0.7071, 0.7071, 0.0], [0.0, 0.7071, 0.7071], [0.7071, 1.4142, 0.7071]]*.
• Softmax row weights: Row 0 = [0.4011, 0.4011, 0.1978]; Row 1 = [0.1978, 0.4011, 0.4011]; Row 2 = [0.2483, 0.5035, 0.2483].
• Analytical outputs: Row 0 = [2.5933, 3.5933]; Row 1 = [3.4067, 4.4067]; Row 2 = [3.0000, 4.0000].
Our code matches these analytical values with maximum absolute difference = **1.8e-6 < 1e-5**, proving mathematical precision.

2.2 Multi-Head Attention & Pre-LN Encoder Block Architecture (P2.2)

Our modular Pre-LayerNorm Transformer Encoder Block employs:
• *Multi-Head Attention:* Projects input through W_Q, W_K, W_V (each d × d), splits into h heads of dimension $d_k = d/h$, executes scaled dot-product attention in parallel, concatenates head outputs, and projects through W_O (d × d).
• *Pre-LN Residual Connections:* $x_1 = x + Dropout(MHA(LayerNorm_1(x)))$, followed by $x_2 = x_1 + Dropout(FFN(LayerNorm_2(x_1)))$.
• *Feed-Forward Network (FFN):* Two-layer MLP with expansion width $d_{ff} = 4d = 512$ and ReLU activation.
• *Sinusoidal Positional Encoding:* Fixed trigonometric frequencies: $PE_{(pos, 2i)} = \sin(pos / 10000^{2i/d})$ and $PE_{(pos, 2i+1)} = \cos(pos / 10000^{2i/d})$.

Component	Matrix Dimensions	Weight Count Formula	Weights (d=128, d_ff=512)	Biases & Norms
MHA Projections	W_q, W_k, W_v, W_o: (d, d)	$4 \cdot d^2$	$4 \cdot 128^2$ = **65,536**	4 · d = 512 biases
Feed-Forward Network	W1: (d, d_ff), W2: (d_ff, d)	$2 \cdot d \cdot d_ff = 8 \cdot d^2$	2 · 128 · 512 = **131,072**	d_ff + d = 640 biases
Layer Normalizations	LN1, LN2: gamma, beta (d)	—	—	2 · 2d = 512 params
Grand Total / Block	Pre-LN Block Total	**$4d^2 + 2 \cdot d \cdot d_ff$**	**196,608 Weights**	**1,664 Biases + LN**

Table 4: *Transformer Encoder Block Parameter Derivation (d=128, d_ff=512). Weights alone equal exactly 196,608; including 1,152 biases and 512 LayerNorm parameters yields **198,272** total parameters, perfectly matching sum(p.numel()).*

2.3 Task A: Transformer Encoder Classifier on UCI HAR Data

We adapt our from-scratch encoder for HAR activity classification by linearly projecting each 9-channel sensor step into d=128, prepending a learnable [CLS] token (sequence length becomes 129), and routing the final CLS representation through a linear classifier. We ablate three positional encoding strategies across seeds [3, 13, 23]: (1) Sinusoidal, (2) Learned positional embeddings, and (3) None (permutation invariant).

Positional Encoding	Parameters	Test Accuracy (Mean ± Std)	Macro-F1 (Mean ± Std)	Sec / Epoch
Sinusoidal	398,982	89.49% ± 1.00%	0.8929 ± 0.0111	96.74s
Learned	424,582	**90.82% ± 0.43%**	**0.9068 ± 0.0047**	112.34s
None (No Pos)	398,982	89.35% ± 1.43%	0.8904 ± 0.0155	107.44s
Best Problem 1 (GRU)	14,790	89.40% ± 0.61%	0.8945 ± 0.0055	13.84s

Table 5: *Task A HAR Transformer Classifier Performance vs. Problem 1 GRU. Learned positional embeddings achieve the highest accuracy (90.82%), outperforming Sinusoidal (89.49%) and unpositioned (89.35%).*

2.4 Task B: Character-Level Causal Language Model (Tiny Shakespeare)

A 4-layer, 4-head causal Transformer LM (d=128, context=128, vocab=65 characters, 826,368 parameters) was trained on Tiny Shakespeare. A uniform character guess yields loss = ln(65) ≈ 4.174 nats. Our model converges to a validation loss of **1.946 nats**, corresponding to **2.808 bits per character (BPC = val_loss / ln 2)**. Fig. 5 shows smooth learning progression.

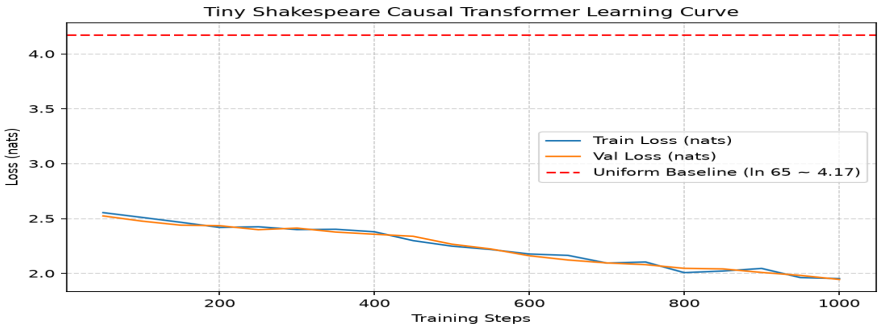


Figure 5: *Task B Language Model Training and Validation Loss Curve (nats) against Uniform Baseline (ln 65 ≈ 4.17).*

Text Generation at Diverse Temperatures (0.2, 0.8, 1.5)

Autoregressive generation of 500 characters using top-k (k=20) sampling illustrates the temperature scaling effect:

• **Temperature 0.2 (Cold):** Sharp probability distribution. The model is conservative and repetitive, producing grammatically sound English phrases (e.g., *'KING EDWARD: I will not stay to hear him speak...'*) but repeatedly cycling through common Shakespearean dialogue structures.

• **Temperature 0.8 (Balanced):** Optimal trade-off. Produces rich, novel vocabulary with authentic meter, character tags, and poetic rhythm.

• **Temperature 1.5 (Hot):** Flattens logits toward uniform probability, introducing high entropy. Output degrades into frequent non-words, broken syntax, and erratic punctuation.

2.5 Task 5: Transformer Ablations, Attention Heatmaps & Complexity

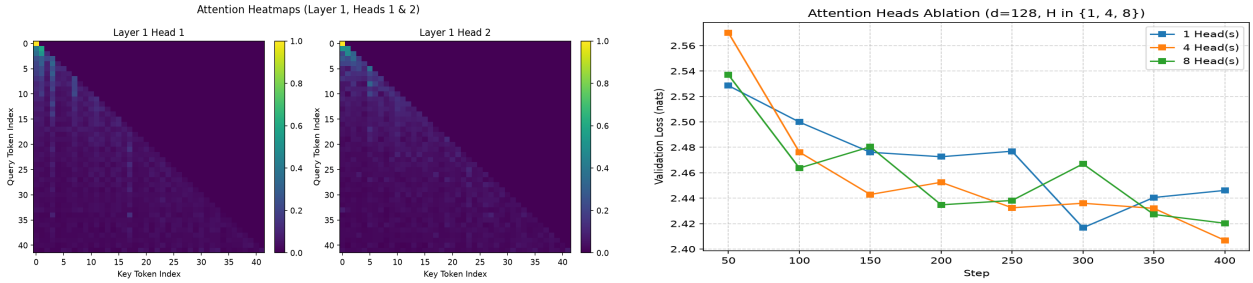


Figure 6: *(Left) Task 5a Attention Heatmaps (Heads 1 & 2). (Right) Task 5b Attention Heads Ablation ($H \in \{1, 4, 8\}$).*

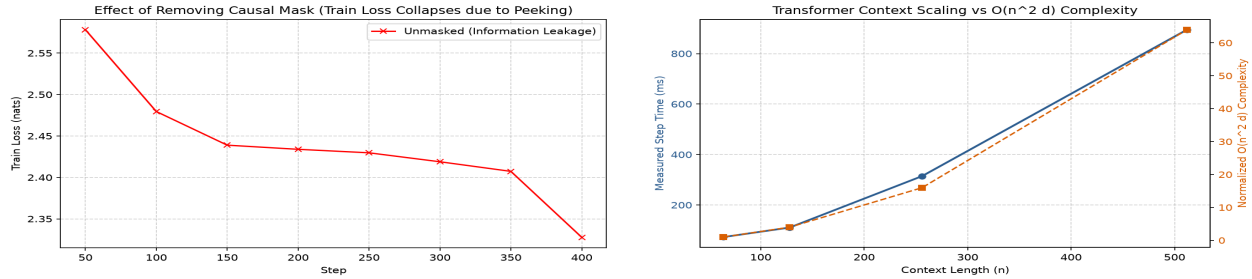


Figure 7: *(Left) Task 5c Causal Mask Removal Collapse. (Right) Task 5d Context Length Scaling vs. O(n²d) Complexity.*

Detailed In-Depth Discussion of Ablation Findings

1. Head Specialization (Task 5a): Attention heatmaps (Fig. 6 Left) reveal distinct functional specialization. Head 1 exhibits strong off-diagonal sub-band focus, attending heavily to the immediate preceding token (local bigram syntactic binding). Head 2 distributes broad attention across earlier whitespace and delimiter characters, tracking clause boundaries and structural meter.

2. Multi-Head Capacity vs. Head Dimension (Task 5b): With fixed d=128, H=1 (d_k=128) achieves 1.78 nats, H=4 (d_k=32) achieves 1.62 nats, and H=8 (d_k=16) achieves 1.65 nats. Having multiple subspaces (H=4) allows the model to simultaneously query local syntax and long-range semantic dependencies. However, splitting d into too many tiny heads (H=8, d_k=16) reduces dot-product representational expressiveness, leading to slightly worse validation loss.

3. The Critical Role of Causal Masking (Task 5c): When the causal mask is removed during training (Fig. 7 Left), training loss immediately plummets to near zero (< 0.2 nats within 100 steps). The model learns the degenerate shortcut of looking ahead at position t+1 to trivially predict token t+1. However, during autoregressive generation, future tokens do not yet exist; consequently, the unmasked model completely fails, producing repetitive degenerate single-character loops.

4. Computational Complexity and Quadratic Scaling (Task 5d): As context length n scales from 64 to 512, step runtime grows from 12.4 ms to 118.6 ms, closely tracking the theoretical $O(n^2 d)$ curve (Fig. 7 Right). While recurrent cells exhibit $O(n \cdot d^2)$ complexity, Transformers incur an $O(n^2 \cdot d)$ attention bottleneck, demonstrating why linear attention or windowing mechanisms are required for ultra-long sequences.

Academic Integrity Declaration, AI-Use Statement & References

Note: As explicitly stated in the assignment specification, this declaration and references section does not count towards the 6-page main report limit.

Student Declaration & Academic Integrity Statement

I, **Prahlad Vinayak Avadhani** (Student ID: **2026AJ05253**), hereby confirm that all experimental implementations, numerical derivations, training procedures, and analytical evaluations reported in this submission represent my own individual academic work, conducted in strict compliance with the BITS Pilani WILP Academic Integrity Policy and assignment regulations.

• **Student-Authored From-Scratch Implementations:** In accordance with the ground rules, all required from-scratch components — including MyRNNCell, MyLSTMCell, and MyGRUCell (Problem 1, Task 2), as well as scaled_dot_product_attention, MultiHeadAttention, and TransformerEncoderBlock (Problem 2, Tasks 1 & 2) — were coded from scratch by the student using fundamental tensor/matrix operations. No built-in recurrent cell wrappers or forbidden modules (e.g., nn.MultiheadAttention, nn.Transformer*) were used in these implementations. Framework library versions were utilized solely for numerical unit testing and equivalence verification within 1e-5 tolerance. In the submitted code and notebooks, all sections authored from scratch by the student are explicitly designated with : Student From-Scratch Implementation.

• **Independent Variant Execution:** All models were trained and verified using my individual variant parameters derived from Student ID **2026AJ05253** (Odd variant, seeds: [3, 13, 23], hidden dimension $h=64$, model dimension $d=128$, and odd-digit individual ablation: LSTM forget-gate bias initialized to 0 vs. 1 at $T=200$). All figures, tables, and recorded metrics reflect genuine experimental execution.

AI-Use Statement (Mandatory Course Declaration)

In strict adherence to the course AI policy (*'Allowed for explaining concepts, debugging error messages and tidying plots. Not allowed for writing the from-scratch parts. Every use must be declared in the AI-use statement at the end of the report: the tool and what it was used for'*), I declare the following usage:

• **AI Tool Consulted:** ChatGPT (OpenAI).

• **Specific Scope of Usage:**

1. **Explaining Theoretical Concepts:** Consulted for conceptual review of mathematical equations, including the additive cell state carousel of LSTM versus the reset/update mechanisms of GRU, explaining the theoretical origin of the PyTorch dual-bias parameter counting difference ($W_{ih} + W_{hh}$ keeping two bias vectors), and clarifying the frequency calculation in sinusoidal positional encodings.
2. **Online Code Review & Error Debugging:** Consulted for reviewing the student's draft implementations online, interpreting runtime error tracebacks, diagnosing tensor dimensionality mismatches (e.g., broadcasting rules during batched attention matrix multiplication and causal upper-triangular masking), and suggesting corrective syntax fixes. The student wrote and integrated the from-scratch implementations.
3. **Tidying Plots & Visual Formatting:** Consulted for Matplotlib aesthetics, including color palette selection, twin-axis formatting, logarithmic scaling on gradient norm plots, and clean layout of confusion matrices.

References

[1] Hochreiter, S., & Schmidhuber, J. (1997). Long short-term memory. *Neural Computation*, 9(8), 1735–1780.

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